

AI for Economic Research: Dynamic Models, Language, and Agents

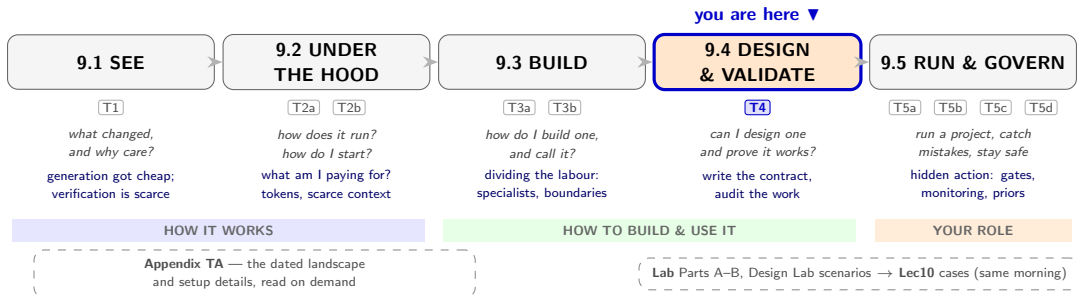
Lecture 9.4: Design Lab — Your Agent for an Economics Task

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Lecture 9 in One Map: Design Lab

This part answers: can you design an agent for a question of your own — and prove it works?



Route: pick a scenario → fill the canvas → write the brief and compile it for your harness → run the detection matrix, three times → pitch what you measured.

The Design Lab in One Slide

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S2 Brief — A Leading Question in a New Costume

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S3 Brief — Fellows Provenance Batch

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The Blank Canvas, and the Brief It Compiles To

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Single-Try Score vs. Reliability Under Repetition

For an agent that runs unattended, the relevant statistic is not its best run — it is whether it succeeds every time.

Best-of- n thinking (wrong for autonomy)

- report the best of several attempts
- flatters the model
- fine for a human-in-the-loop draft you will check

pass-hat- k / reliability (right for autonomy)

- probability *all* k independent runs succeed
- τ -bench: $\sim 90\%$ at $k=1 \rightarrow \sim 57\%$ at $k=8$
- this is what matters when no human checks each run

Economist translation: an unattended agent on a recurring data pull is a repeated trial. A 90% single-try success compounds into frequent failure across a quarter of nightly runs. Design for the reliability curve, not the demo.

Pitch and Validate: The Detection Matrix at Rung 3

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Bridge: From One Agent to a Whole Project

You designed one agent for one task and measured whether it catches what it should.

- **Next (Part 9.5):** run a whole project with a team of them (T5a–T5b), catch your own mistakes (T5c), and stay inside the red lines (T5d)